The Effects of Restricted Abortion Access on IUDs and Vasectomies: Evidence from Texas

Bradley Crowe[‡] Graham Gardner[†] Cara Haughey[‡]

Abstract

Both contraception and abortion result in fertility reductions, but whether they are substitutes remains an open question. In 2013, Texas passed House Bill 2 (HB2), a policy that imposed strict regulations on abortion providers. Using administrative outpatient records from Texas, we exploit the passage of HB2 to identify the effects of restricted abortion access on the timing and demand for intrauterine devices (IUDs) and vasectomies using an event study design. We find evidence that expectations of limited abortion access significantly increase the demand for IUDs, with no effect on the incidence of vasectomies. These findings support the hypothesis that abortion and contraception are substitutes, particularly for individuals with the capacity to become pregnant.

Keywords: abortion, family planning, TRAP laws, contraception, long-acting contraception, IUD, vasectomy
JEL Codes: I10, I18, J13, J16

[‡] Michigan State University, Department of Economics
[†] Texas Christian University, Department of Economics

1. Introduction

In 2022, the United States Supreme Court transferred the power to regulate abortion access to individual states in the landmark case *Dobbs v. Jackson Women's Health Organization*. This decision ultimately ruled that the U.S. Constitution does not guarantee the right to abortion, overturning *Roe v. Wade* (1973) and *Planned Parenthood v. Casey* (1992). Prior to the ruling, thirteen states enacted “trigger" laws to ban or limit abortion immediately after the repeal of *Roe*. Many states created laws to restrict abortion access or prohibit abortion entirely in the months that followed the *Dobbs* decision. In the 100 days following *Dobbs*, 66 abortion clinics closed across 15 states (Kirstein et al., 2022), creating substantial barriers to abortion access in the United States.

Limited access to abortion could affect fertility and sexual behavior through multiple channels. For example, reduced access to abortion may lead to increases in births, particularly for those who would have chosen abortion if the option were available. Alternatively, restricted abortion access may lead to fewer births through either decreases in sexual activity or increases in contraceptive use. Prior research finds that restricted abortion access affects birth rates (Jones & Pineda-Torres, 2024; Venator & Fletcher, 2021; Myers & Ladd, 2020; Guldi, 2008; Kane & Staiger, 1996) and sexual behavior (Colman et al., 2013; Klick et al., 2012). However, research on how changes in abortion access affect contraceptive use is limited.

In this paper, we begin to disentangle the relationship between abortion access and the use of intrauterine devices (IUDs) and vasectomies. If abortion access and contraceptives are substitute goods, then economic theory suggests that increasing the cost of accessing an abortion will increase the demand for contraceptives. Much of the literature shows that changes in abortion access increase fertility and birth rates (Jones & Pineda-Torres, 2024; Venator & Fletcher, 2021;

Myers & Ladd, 2020). However, if substitution behavior between contraceptives and abortion is not accounted for, these studies may be underestimating the effects of abortion on fertility. Given that restrictions on contraceptive access are often targeted alongside abortion restrictions (Felix et al., 2024), identifying the extent of substitution behavior between abortion and contraceptives can help predict effects on fertility when laws restricting both types of family planning are passed concurrently. In addition, understanding the substitutability between abortion and contraception, particularly post-*Dobbs*, can provide policymakers with strategies to support individuals as the cost of abortion increases. Considering the evidence that limiting abortion access increases maternal mortality and morbidity (Gardner, 2024; Kheyfets et al., 2023), providing affordable access to contraception may mitigate the effects of restricted abortion access on maternal health outcomes.

To identify the relationship between abortion access and contraceptive behavior, we leverage a natural experiment arising from changes in abortion policy in Texas. In 2012, a series of highly publicized political developments contributed to a politically charged, anti-abortion climate that culminated in the passage of House Bill 2 (HB2). This legislation imposed stringent regulations on abortion providers, leading to the closure of more than half of the state's clinics. We exploit both the threat of these closures and the resulting variation in distance to the nearest abortion provider to examine whether individuals altered their contraceptive behavior in response. Using administrative outpatient records from hospitals, hospital-affiliated facilities, and ambulatory surgical centers (ASCs), we analyze trends in the incidence of intrauterine devices (IUDs) and vasectomies.

We find that overall hospital-based IUD[1] insertions rose in Texas between 2011 and 2015, and trends peak around the introduction and passage of HB2. In counties that experience a greater-than-30-mile increase in their distance to an abortion provider, IUD incidence increases at a significantly higher rate around the time of the policy change. However, we find little evidence that restrictions to abortion access change the rate of vasectomy procedures.

This paper contributes to the expanding literature on the effects of restrictive abortion and contraceptive policies, which finds detrimental socioeconomic and health outcomes resulting from decreased access to abortion, in addition to the aforementioned effects on fertility and sexual behavior. Previous studies find that abortion restrictions cause poor pregnancy-related health outcomes, decreases in college-going behavior, and decreases in future family income (S. Miller et al., 2023; Farin et al., 2021).

Other work specifically evaluates the effects of abortion restrictions due to HB2 and other fertility related legislation in Texas. For example, Lindo et al. (2020) finds that changes in the distance to the nearest abortion provider due to HB2 reduced abortion rates and increased birth rates within the state. Stevenson et al. (2016) evaluate the effect of separate Texas legislation that excluded Planned Parenthood from Medicaid reimbursement and finds decreases in long-acting reversible contraceptives (LARC) and injectable contraceptive utilization among Medicaid recipients. In a paper that also looks responses to HB2, Fischer et al. (2018) analyze the effects of HB2 on abortion and birth rates. The authors find a 1.3% increase in births and a 16.7% decrease in abortions in counties that lost access to an abortion provider within 50 miles due to HB2. The authors also find no effect on retail contraceptive use measured by emergency

[1] Long-acting reversible contraceptives (LARC) include hormonal and non-hormonal IUDs and subdermal contraceptive implants. We observe very few subdermal implants in our sample and therefore do not look at the effect of HB2 on subdermal implants in this paper.

contraceptive and condom sales. Our work complements this research and provides a possible explanation for why Fisher et al. (2018) find no response in retail contraceptive use; perhaps substitution behavior in this setting primarily leads to increases in IUDs, rather than retail contraceptives, suggesting that individuals may prefer contraceptive methods that are better at preventing pregnancy when facing abortion restrictions. Research by Pennington & Venator (2024) supports this theory, showing that women switch to more effective forms of birth control in response to abortion restrictions.

We also add to the literature on contraceptive use and family planning. Much of the previous research uses increased access to contraceptives to examine how contraceptives decrease family size and improve educational and labor market outcomes for women by giving them more control over the timing and frequency of births (Bailey, 2005, 2010, 2013; Goldin & Katz, 2002). Further, researchers find that access to contraceptives leads to a decrease in the share of children born to economically disadvantaged households and a decrease in the number of children with low birthweight (Ananat & Hungerman, 2012). However, much of this literature focuses specifically on one type of contraceptive: the oral contraceptive pill. Our analysis adds to this work by studying the use of IUDs and vasectomy, both of which provide more effective protection against pregnancy.

Lindo & Packham (2017, 2020) examine LARC utilization using expanded access to LARC rather than contracted access to abortion as their main source of variation. The authors find that Colorado's Family Planning Initiative, which invested in expanding LARC access, reduced birth rates by 20 percent for women living within 7 miles of a Title X clinic when compared to women living farther away (Lindo & Packham, 2020). The authors find suggestive evidence that the initiative reduced unintended pregnancies that would have ended in abortion, particularly among teenagers. Combining the results from Lindo & Packham (2020) and our findings suggests that expanding access to LARC in response to abortion restrictions may offer protection against unwanted pregnancies, particularly for young adults and teenagers.

Finally, we complement other work evaluating the relationship between abortion access and contraceptive choices. G. Miller & Valente (2016) use health surveys to understand how the rapid expansion of legal abortion access in Nepal changed contraceptive utilization. The authors find evidence that an additional abortion clinic in an individual's district decreases the odds of contraceptive use by 2.6 percent. In the United States context, however, the evidence of the relationship between contraception and abortion is less clear. Felkey & Lybecker (2017) find that state-level abortion restrictions do not affect the reported contraceptive method of choice among reproductive-age women. On the other hand, Ellison et al. (2024) find descriptive evidence of an increase in permanent contraceptive choices, such as tubal ligation and vasectomy, after the *Dobbs* decision.

The remainder of this paper proceeds as follows. Section 2 explains the relevant history and background, including a description of IUDs and vasectomies and the policy environment that generates our source of variation. Section 3 describes the data and Section 4 presents our methods and results. Section 5 concludes.

2. History and Background

2.1. Long-Acting Reversible Contraception and Vasectomy

Contraceptive users choose from several different contraceptive options, all with different efficacy rates, financial costs, and side effects.[2] IUDs are one type of long acting and highly

[2] Other types of contraceptive methods include the contraceptive pill, shot, patch, and vaginal ring, condoms, diaphragm, sponge, spermicide, fertility awareness, withdrawal, and abstinence. For complete details on contraceptive methods, see https://www.plannedparenthood.org/learn/birth-control (Planned Parenthood, 2024).

effective contraceptives. IUDs are flexible, T-shaped devices placed in the uterus by a physician. The device prevents fertilization by decreasing sperm motility using either hormonal or non-hormonal properties, depending on the type of IUD. The non-hormonal IUDs, such as Paragard, are wrapped in copper and can protect from pregnancy up to 12 years. Hormonal IUDs, including Mirena and Kyleena, can protect from pregnancy between 3 and 8 years, depending on the product.

Although IUDs were introduced shortly after the oral contraceptive pill, they received significant negative press following a series of studies in 1974 that linked IUDs to pelvic inflammatory disease, negatively impacting their popularity (Sonfield, 2015). Modern IUDs were introduced in 1988 (Paragard) and 2001 (Mirena). Initially, IUDs were only approved by the Food and Drug Administration (FDA) for use by people who had given birth, however the FDA approved IUDs for people who have never give birth in 2005. In spite of the updated FDA guidelines, misconceptions about the safety of IUDs for people who have never given birth remained prevalent for years after the updates (Tyler et al., 2012). Table 1 lists the various types of contraceptives and their use frequency among users between 15 and 49. While IUDs are among the most popular contraceptives, with 12.9 percent of contraceptive users choosing IUDs, use remains lower than oral contraceptives, with 21.4 percent of contraceptive users choosing oral contraceptives.

The ease of use and effectiveness of IUDs may make them particularly attractive to individuals facing increased barriers to abortion access. IUDs are over 99 percent effective at preventing pregnancy with typical use. In contrast, condoms are 87 percent effective, and the oral contraceptive pill is 93 percent effective at preventing pregnancy with typical use. Further, while oral contraceptives and condoms are subject to user error, IUDs are inserted and removed in a physician's office and therefore require very little effort on the part of the user. However, IUD insertion and removal can be painful, and users may experience unpleasant side effects including irregular menstrual bleeding, spotting, and mood changes (Lindo & Packham, 2017)

Table 1: Contraceptive method choice
Most effective method used in the past month by U.S. women, 2018

Method	% of contraceptive users, 15-49
Permanent Female	27.7
Pill	21.4
Male condom	12.9
IUD	12.9
Partner Vasectomy	8.6
Withdrawal	5.6
Implant	3.1
Injectable	3.0
Natural family planning	2.6
Vaginal ring	1.3
Patch	0.5
Emergency contraception	0.2
Other method	0.3
No method	n/a
Observations	47,444,047

Source: Contraceptive Use in the United States by Method; Guttmacher Institute, 2021

Vasectomy is an outpatient, surgical procedure that can be completed under local anesthesia. This method of contraception is done by puncturing or severing the vas deferens, which is the tube that carries sperm. It is nearly 100 percent effective as a form of contraception, though pregnancy can occur shortly after the procedure, before sperm are cleared from the reproductive tract (Schwingl & Guess, 2000). Vasectomy is not meant to be reversed, although reversal is possible in some cases and up to six percent of vasectomized people request a reversal

(Dickey et al., 2015). As with most surgical procedures, a vasectomy is not without risk of complication and side effects, although these risks are small.

Side effects are one potential reason that users may choose another form of contraception over an IUD or vasectomy. Additionally, IUD users may face higher upfront costs than other contraceptive choices. For example, IUDs cost between $760 and $1,101 for the device alone, exclusive of the cost of insertion. Similarly, vasectomies can cost around $1,000 (*Planned Parenthood*, n.d.). However, in 2012, the Affordable Care Act (ACA) mandated that insurance plans cover birth control, including IUDs and vasectomy, without any out-of-pocket costs to the patient. The ACA mandate was effective, and Bearak et al. (2016) show that while 58 percent of women faced some out-of-pocket costs for contraceptives prior to the mandate, that number dropped to 13 percent by March of 2013. Further, the authors find that the cost estimates also declined; women facing the 90th percentile cost prior to the ACA were expected to pay around $844 during the time of the study, and by January 2013, the 90th percentile cost estimate declined to $169. Texas implemented HB2, the policy studied in this paper, in July of 2013, and therefore individuals were likely facing low- or no-cost for their procedures during this time period.

2.2. Legislative Framework

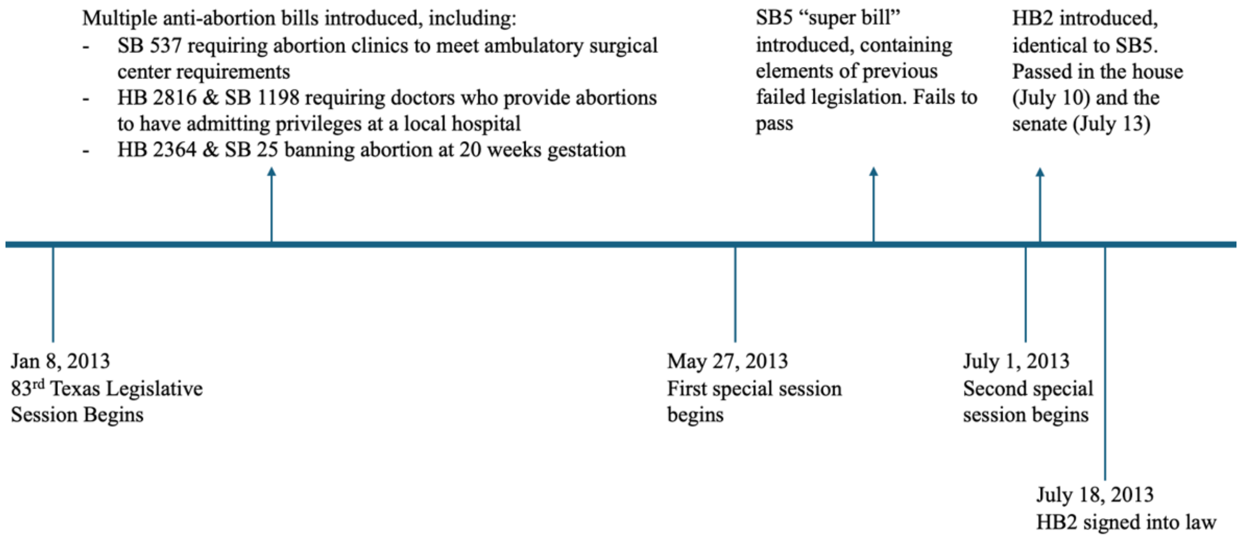
The 2013 efforts to restrict abortion access that resulted in HB2 followed years of budget cuts and other efforts that targeted family planning clinics. In 2011, the Texas government reduced funding for family planning clinics in the 2012-2013 budget by 67% (Fischer et al.; 2018). In response, 25 percent of family planning clinics in the state closed in 2012 and other clinics reduced services by limiting hours or eliminating staff (White et al., 2015). Additionally, the Texas government replaced the federal Women's Health Program, a program that provides subsidized women's healthcare for low-income women, with the Texas Women's Health Program. Shifting from a federally funded to a state funded program allowed the state of Texas to exclude clinics affiliated with an abortion provider from receiving reimbursement through the Women's Health Program. This change passed in 2011 and went into effect in January of 2013.

The tense, anti-abortion environment in Texas was tracked carefully by the media, perhaps beginning with a highly publicized speech by Governor Rick Perry in December 2012 stating that his goal was to "make abortion at any stage a thing of the past." This speech made national headlines (CBS News, 2012; Marcotte, 2012; Bassett, 2012) and was covered in prominent Texas publications (Hopkins, 2012; Ramshaw, 2012).

HB2 passed after multiple abortion bills were introduced and failed during the regular legislative session. Governor Perry ordered a special session to begin in late May 2013. During this session, the Texas Senate introduced Senate Bill 5 (SB5), a bill containing many details of the abortion bills that failed to pass in the regular session. When SB5 failed to pass, Governor Perry instituted a second special session to begin in July. SB5 was introduced again as HB2, which passed both houses and was signed into law on July 18, 2013. Figure 1 shows the timeline of events leading to HB2.

Those who supported the bill claimed it would protect women's health by 1) requiring doctors performing abortions to have admitting privileges at a hospital within 30 miles of the clinic, 2) ensuring individuals taking abortion-inducing medication have medical oversight, and 3) mandating that abortion providers meet the standards of an ambulatory surgical center. Opponents of the bill argued that these restrictions, known as targeted regulations on abortion providers, or TRAP laws, would unnecessarily burden clinics to the point of closure, thereby restricting access to legal abortion (Fernandez, 2013).

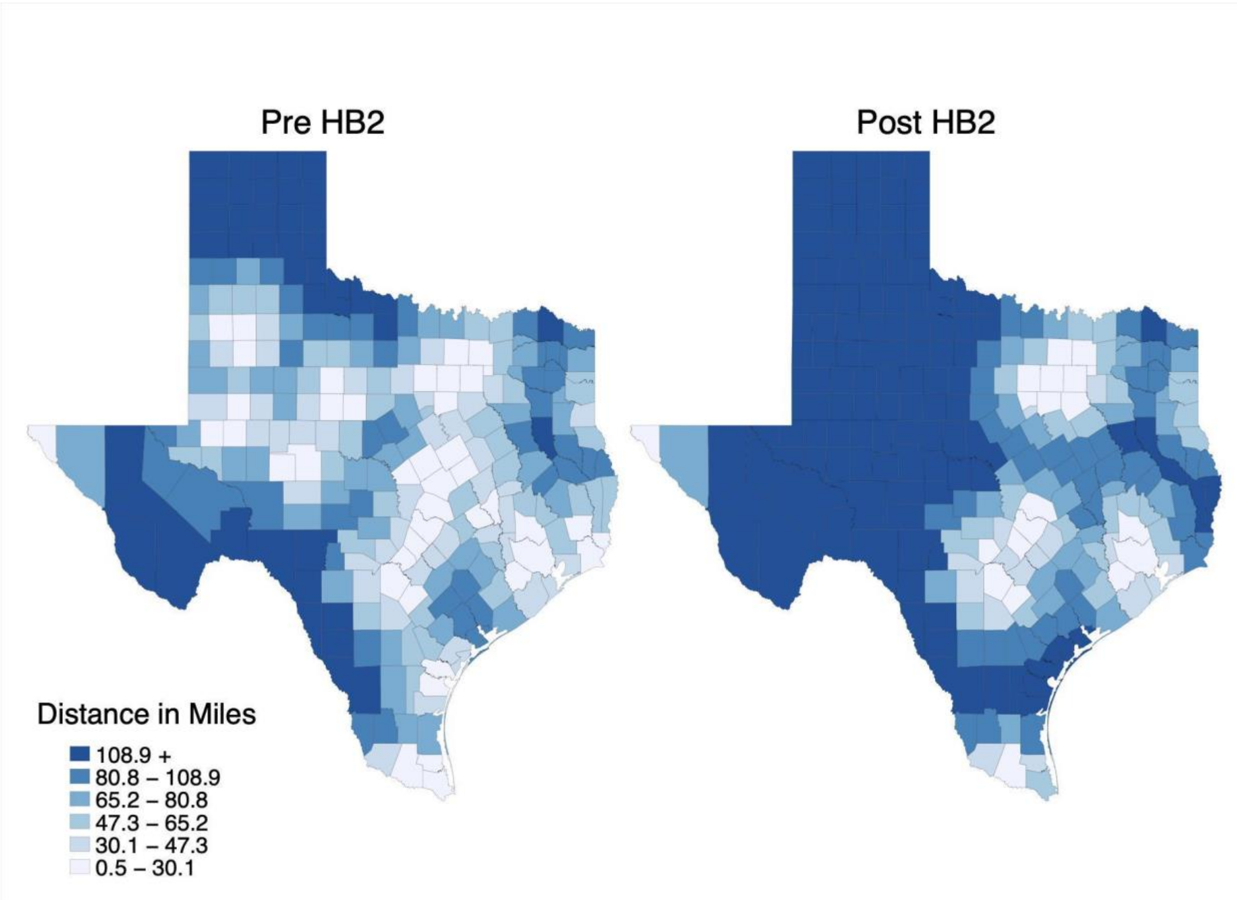
Figure 1: House Bill 2 Legislative Timeline



The first and second mandates of the law went into effect in November of 2013, and the number of abortion clinics in Texas dropped significantly.[3] In 2011, Texas had 42 operating abortion facilities and by the end of 2014, only 17 remained. Before HB2, a majority of Texas residents lived within 100 miles of an abortion provider, and many lived within 30 miles of a provider. After the implementation of HB2, clinic access was concentrated in the metropolitan areas, and many residents faced a journey of over 100 miles to access abortion care, including travel to other states. This change in abortion clinic access not only increased the travel burden for individuals seeking an abortion but also resulted in congestion and long wait times at the remaining clinics (Lindo et al., 2020).

[3] The mandate that abortion providers meet the standards of an ambulatory surgical center went into effect in the fall of 2014.

Figure 2: Average travel distance to an abortion provider in Texas



Notes: The average travel distance to an abortion provider is calculated at the county-level using Myers (2021). The figure on the left shows the average county-level travel distance to an abortion provider before the introduction of HB2 in June 2013 and the figure on the right shows the average travel distance after HB2. The darker blue represents longer distances.

The left panel of Figure 2 shows the average travel distance residents of a particular county faced to access their closest abortion provider prior to the passage of HB2. Individuals in the lightest shaded counties had an average travel distance of less than 30 miles and those in the darkest shaded counties had an average travel distance of at least 109 miles. The right panel shows the average travel distance after the passage of HB2. Much of west and central Texas lost access to a close abortion provider, as shown by the shift in color from light to dark.

3. Data and Descriptive Statistics

To measure changes in the incidence of IUDs and vasectomies in Texas, we use quarterly data from the 2011-2015 Texas Health Care Information Collective Outpatient Public Use Data Files (THCIC, 2011-2015). THCIC collects healthcare claims, or invoices submitted to health insurance companies, reported by Texas hospitals and makes the de-identified data available for public use. The data include rich details on each patient claim, such as diagnosis and procedure codes associated with the claim, the timing of the claim at the quarter-year level, and patient demographic information, such as gender, race, and county of residence. All state licensed hospitals, hospital-owned outpatient facilities, and ambulatory surgical centers within the state of Texas are required to report claims, representing 782-956 facilities in each quarter-year.[4] We identify 8,379 IUD insertions or IUD removals with reinsertion and 6,314 vasectomy procedures using relevant Current Procedural Terminology (CPT) codes.

Table 2 shows differences in some demographic features of IUD users and vasectomy recipients in our sample. IUD users are generally young, with a mean age range of 25 to 29, while vasectomy recipients average between 35 and 39 years old. The primary payer for IUDs is overwhelmingly Medicaid, which typically covers low-income individuals. On the other hand, Medicaid pays for less than 1% of vasectomy procedures. Instead, commercial insurance providers cover vasectomies in the data. Therefore, while IUD users skew younger and likely lower-income in our sample, vasectomy users tend to be older and likely higher-income relative

[4] Small, rural hospitals, such as those located in counties with a population of less than 35,000 or those with fewer than 100 hospital beds are not required to report. Additionally, facilities that do not seek insurance or government reimbursement are not required to report. Because facilities come in and out of our sample during the study period, we perform a robustness check in Appendix A6 using a balanced sample of facilities.

to IUD users. Given these differences, along with differences in the benefits and efficacy of the procedures, we expect a heterogeneous response in IUD and vasectomy choice following HB2.

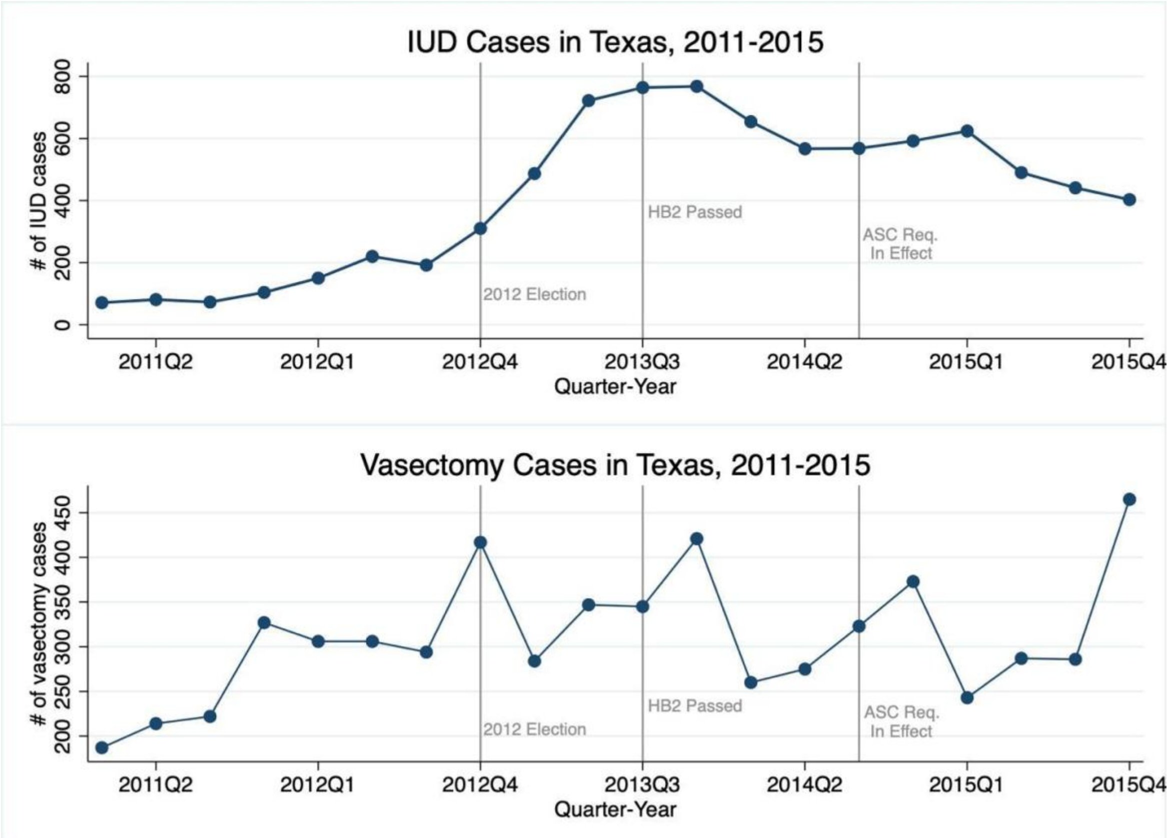
Table 2: Summary Statistics of IUD and Vasectomy Recipients in Texas, 2011-2015

	IUD		**Vasectomy**	
Variable	**Mean**	**N**	**Mean**	**N**
Age (year category)	25-29	8379	35-39	6314
Race (%)		8379		6309
White	47.50		57.30	
Black	13.78		4.99	
AAPI	3.22		0.87	
American Indian	0.16		0.17	
Other/Unknown	35.34		36.63	
Ethnicity (%)		8371		6290
Hispanic	39.01		17.74	
Non-Hispanic	60.99		82.26	
First Payment Source (%)		8378		6290
Medicaid	25.90		0.52	
Charity, Indigent, or Unknown	27.51		5.78	
Health Maintenance Org.	11.02		8.82	
Commercial Insurance	10.36		37.86	
Blue Cross/Blue Shield	9.41		17.54	
PPO	9.04		23.13	
Other	6.76		6.35	

Source: THCIC (2011-2015)

Figure 3 demonstrates how the incidence of IUDs and vasectomies evolved over the study period. The raw trends in IUD cases started increasing around the time of the 2012 election, coinciding with Governor Perry's widely reported speech in the fourth quarter of 2012. Following the 2012 election, the balance of power shifted in the Texas legislature with a small Republican majority of 2 votes switching to a large, 48-vote majority. The comfortable majority in the legislature motivated anti-abortion groups to push for additional abortion restrictions in 2013 (Fernandez & Eckholm, 2012).

Figure 3: Trends in IUDs and Vasectomies



Notes*: The top figure shows the count of IUD insertions between 2011 and 2015, and the bottom figure shows the count of vasectomy cases for the same period. Each blue point represents the count of cases in the quarter relative to the passage of HB2 in July 2013, which is represented by the grey line. Additionally, we include a line for the date of the 2012 election and a line for September 2014, when the ambulatory surgical center requirement went into effect.*

Changing contraceptive behavior in response to Governor Perry's rhetoric and shifting political power is consistent with evidence of 'defensive investments' in contraception described in Pennington & Venator (2024). The authors show that people receiving care in Planned Parenthood facilities in Wisconsin switched to more effective contraceptive methods after the governor proposed new abortion restrictions but before the restrictions were even introduced in the legislature. Responding to non-price factors, such as media coverage, has also been documented in response to cigarette taxes (Rees-Jones & Rozema, 2023). Forward-looking agents may use their perception of expected future abortion access in their present contraceptive decisions, and therefore we define our time of treatment as the passage of HB2 in the third quarter of 2013.

In the time leading up to the introduction of HB2, the number of IUD insertions increased from around 200 to nearly 800 per quarter. A more modest increase in IUDs occurred following enforcement of the ambulatory surgical center requirement in September 2014. After the policy changes, the incidence of IUDs declines back toward pre-treatment levels, consistent with the hypothesis that people respond to new information about the availability of abortion when making contraceptive choices. Given that IUDs are long-lasting forms of contraceptives, we do not expect the changes in the incidence of IUDs to persist continuously over time.

Trends in the incidence of vasectomy are relatively noisy, which appear to be driven by strong seasonal effects. Seasonality in vasectomy procedures is well known; Ostrowski et al. (2018) find that vasectomy procedures peak at the end of the year when patients have met their insurance deductible. However, even outside of the seasonal effects, it does not appear that these trends are strongly correlated with the timing of the abortion legislation.

To determine a measure of treatment from HB2, we use a panel describing the distance to the nearest abortion provider from Myers (2021). Myers (2021) calculates the travel distance in miles from the county population centroid to the nearest abortion providing facility in each month from 2009 to 2021. We average these distances across quarter-years and then match the aggregated distance measures to our outcome data for Texas counties from quarter one of 2011 to quarter four of 2015. A county is treated if the distance to the closest abortion provider increases by more than 30 miles.[5] Counties with a change that is less than 30 miles comprise the

[5] We choose a 30-mile cutoff because it is equivalent to the 30-mile hospital distance regulation imposed by HB2. We test alternative cutoffs in Appendix A2.

control group. This definition of treatment results in 117 treated counties and 137 control counties.

Table 3: Summary Statistics of IUD and Vasectomy Treatment and Control Groups, 2011-2015

	IUD		**Vasectomy**	
Variable	**Control**	**Treated**	**Control**	**Treated**
Age Category (%)				
18-19	4.24	3.76	0.04	0.00
20-24	18.15	17.00	0.73	1.55
25-29	20.60	21.29	5.50	10.98
30-34	19.19	20.76	20.11	28.16
35-39	14.86	15.71	28.20	27.38
40-44	11.19	11.12	23.93	16.08
45-49	6.02	5.82	12.38	9.53
Race (%)				
White	41.97	67.65	56.69	61.75
Black	14.28	11.88	5.15	4.32
AAPI	3.76	1.12	0.90	0.78
American Indian	0.15	0.06	0.15	0.33
Other/Unknown	39.83	19.29	37.11	32.82
Ethnicity (%)				
Hispanic	45.90	13.53	18.20	15.08
Non-Hispanic	54.10	86.47	81.80	84.92
First Payment Source (%)				
Medicaid	29.89	11.12	0.55	0.44
Charity, Indigent, or Unknown	34.71	1.12	5.95	5.21
Health Maintenance Org.	6.27	29.65	6.79	21.29
Commercial Insurance	7.02	22.18	39.02	29.49
Blue Cross/Blue Shield	5.75	23.12	15.34	28.60
PPO	10.89	1.59	26.37	5.99
Other	5.47	11.24	5.97	8.98
N (Patient-Quarters)	6,540	1,700	5,241	902

Source: THCIC (2011-2015)

Table 3 shows summary statistics for IUD and vasectomy patients, comparing those in the control group and the treated group. Patients in the IUD control group tend to be younger, less likely to be white, and more likely to be Hispanic than those in the treated group. Additionally, over half of the patients in the IUD control group use either Medicaid or charity, indigent, or unknown sources for their payment, but most in the treated group use commercial, Blue Cross Blue Shield (BCBS), or HMO health plans. Most patients in the vasectomy cohort rely on HMO, PPO, BCBS or other commercial insurance plans, with the control group relying slightly more on commercial insurance.

Figure 4 plots raw trends in the rate of IUD insertion and the rate of vasectomies[6] for treatment and control counties. The figure indicates that rates of IUD insertion increase in both the treatment and control group following the 2012 election, consistent with the hypothesis that individuals may be engaging in defensive investments in contraception. Even before HB2 is passed, however, IUD rates increase more heavily in treated counties. HB2 imposed clear restrictions on abortion providers, such as mandating admitting privileges at a hospital within 30 miles and requiring clinics to meet the standards of an ambulatory surgical center. News articles circulated in the Texas Tribune and NBC News, including details regarding which clinics might be affected by proposed requirements (Ramshaw, 2013; NBC News, 2013). This supports the theory that Texas residents understood which clinics were most likely to close following implementation of HB2, leading to behavior shifts before HB2 was implemented.

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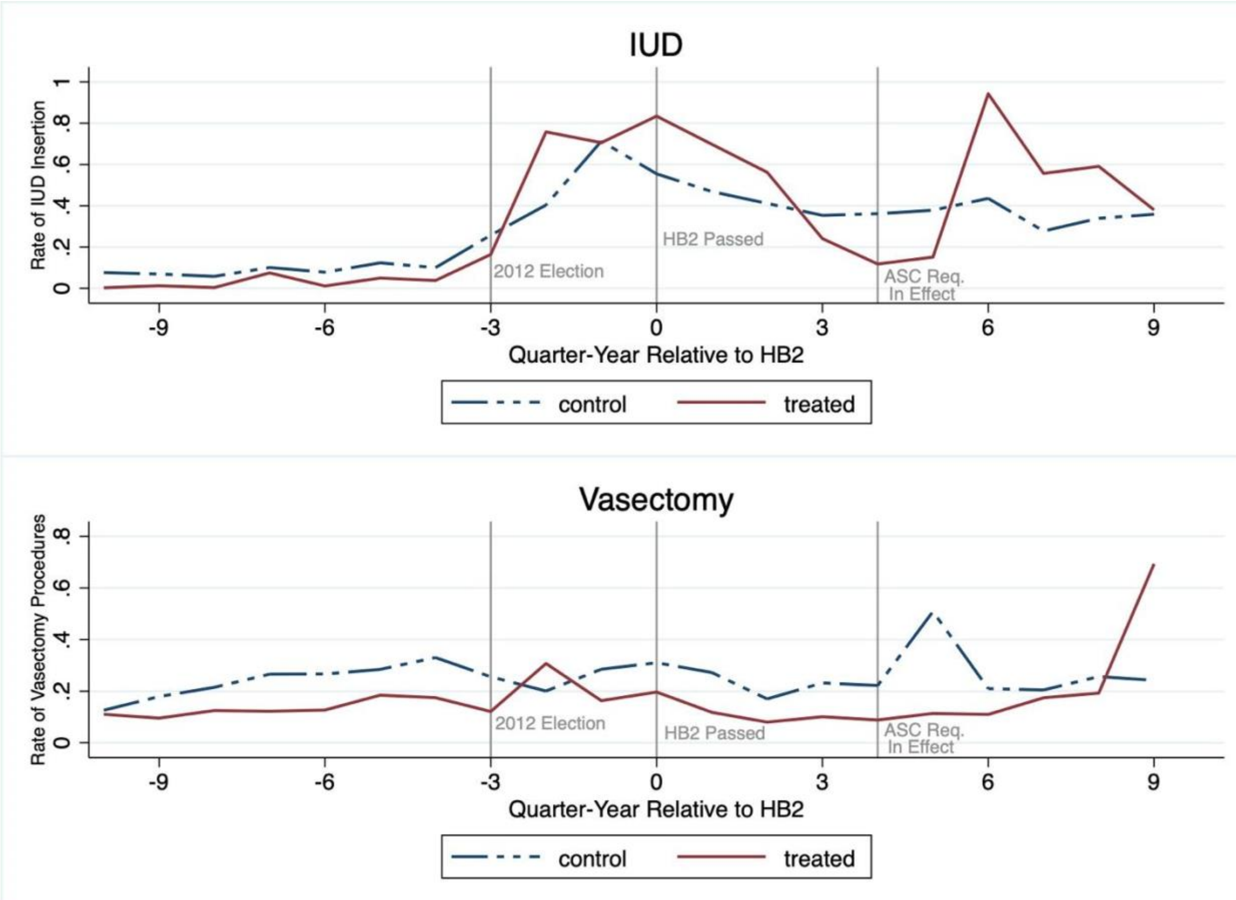
[6] Rates of IUD insertion are defined per 10,000 females between 15-49 and rate of vasectomies is defined per 10,000 adult males.

Figure 4: Trends in IUDs and Vasectomies by Treatment Status



Notes: The top figure depicts trends in the rate of IUD insertions per 10,000 females aged 15-49 relative to July 2013, when HB2 was passed. The bottom figure depicts trends in the rate of vasectomies per 10,000 adult male. The red trend line shows trends in the treated counties, while the blue dashed line shows the trends in the control counties. Treatment is defined as counties that experience an increase in the distance to an abortion provider of more than 30 miles following HB2.

After HB2 passed, IUD rates began to decline but remained higher in treated counties for nine months after the policy change. Soon after the ASC requirement went into effect in September 2014, IUD rates spiked in treated counties, with little-to-no change in the control counties. Trends in the rate of vasectomies are largely similar across the treatment and control group, with occasional spikes that do not appear to be closely associated with the policy change.

4. Method and Results

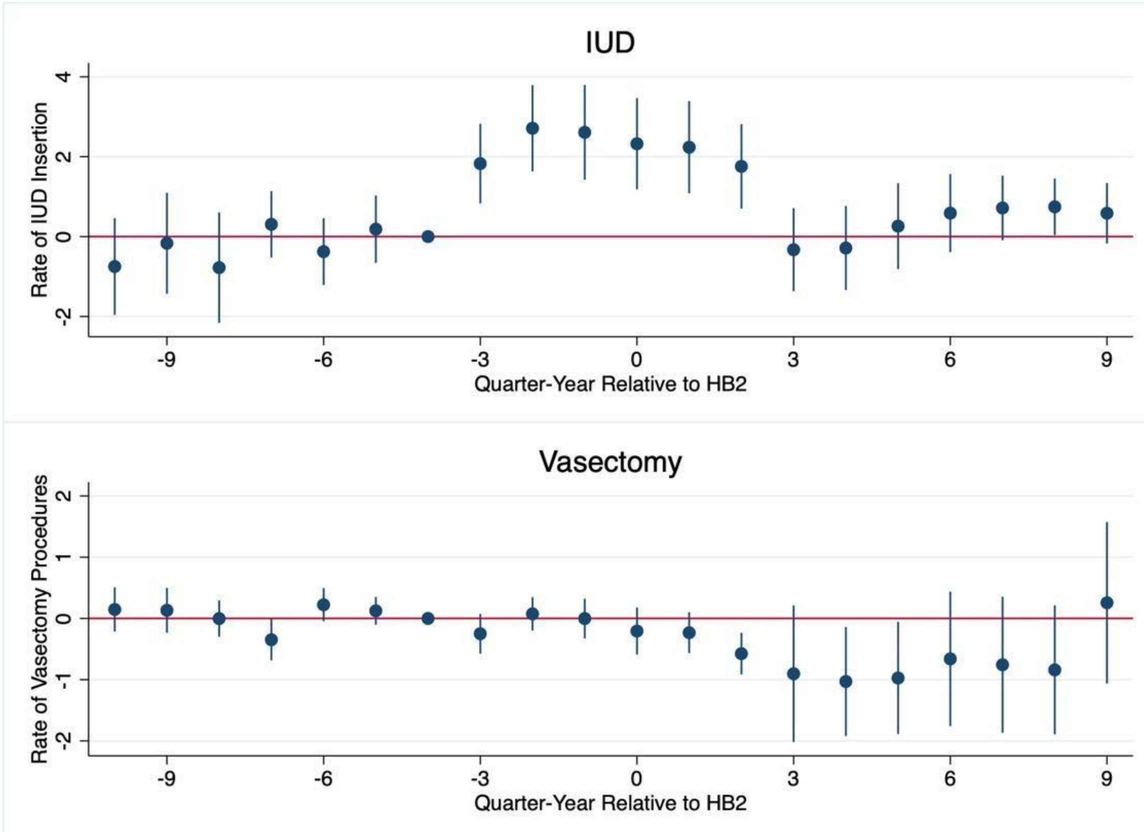
4.1. Main specification

To pinpoint the magnitude of the effect of restrictive abortion policies on IUD and vasectomy choice, we rely on an event-study design. The specification in Equation (1) captures the anticipation effects we expect when people are making defensive investments in contraception. Because the data contain discrete counts of IUD and vasectomy incidence occasionally equal to zero, we employ a Poisson model that takes the form:

$$E[Y_{ct}|\alpha_c, \delta_t, \sum_{k=-10}^{9} \lambda_k 1(t = k)] = \exp(\alpha_c + \delta_t + \sum_{k=-10}^{9} \lambda_k 1(t = k)) \qquad (1)$$

where Y_{ct} is the number of IUDs or vasectomies in county c at quarter-year t, and α_c and δ_t are county and quarter-year fixed effects respectively. Here, k indexes the coefficients λ relative to the passage of HB2 in the third quarter of 2013. We include an exposure variable for the total number of females between 15 and 49 (IUD) or adult males (vasectomies) for county c in quarter-year t. In doing so, we interpret our results as the effect of reduced abortion access on the rate of IUD insertions or vasectomy procedures. Our identifying assumption is that counties in the treated group would have trended similarly to counties in the control group, absent the policy intervention.

Figure 5: Poisson Event Studies



Notes: Figure 5 plots the λ_k estimates from Equation (1). Zero represents the passage of HB2 in July 2013, and each point represents differences in the rates of IUD or vasectomy procedures between the treated and control counties relative to four quarters before HB2. This reference point represents the quarter-year prior to the 2012 election.

Figure 5 presents the Poisson event-study plots of λ_k for each contraceptive method along with the 95 percent confidence intervals. For IUD insertions, the event plot reveals that trends are parallel between treatment and control counties leading up to three quarters prior to the introduction of HB2. In the three quarters leading up to HB2, the rate of IUD insertions increased significantly in treated counties. Notably, this timing coincides with the media coverage of the 2012 state elections and the 83rd legislative session, which is likely driving the anticipation effects. Treated counties have a higher rate of IUD insertion until three quarters following the policy change, at which point trends return to close to pre-treatment levels.

For the three quarters before and after HB2 passed, the rate of IUD insertion increased 9.04x more in the treated counties compared to the control counties. While this effect size seems large, it is mostly driven by low baseline rates of IUD insertion in the treated counties. The effect is equivalent to an additional 0.55 IUD insertions per 10,000 reproductive age females in each treated county-quarter, or a total of 326 additional IUD insertions over the study period. Though our paper is the first to consider how restricted abortion access impacts IUD take-up, Kelley et al. (2020) find similarly large estimates in response to expanded LARC access through Title X clinics. The authors estimate that improved access to LARC increased LARC utilization among women under 30 from 3-7 percent to nearly 30 percent.

In the event plot for vasectomies, trends are parallel leading up to the time of the policy change, but there is not strong evidence[7] of an effect for treated counties after the introduction of HB2. The results do indicate a small negative effect in 2014, which could be explained by the stable trend in vasectomies in treated counties compared to the general increase among the control group, depicted in Figure 4.

4.2. Responding population

Our primary results point to increases in IUD utilization across the population after the implementation of HB2. However, Sutton et al. (2021) discuss persistent inequities and disparities among racial and ethnic minorities, potentially driven by implicit bias by healthcare workers, patient mistrust of the healthcare system, differential access to health insurance, high upfront costs of contraceptive care, and other barriers to access. In addition, critics of abortion

[7] While we do not see any effects for vasectomy procedures, it is plausible that individuals seek out vasectomy consultations and ultimately decide against the procedure. In light of recent evidence that shows significant increases in demand for vasectomies after *Dobbs* (Ellison et al., 2024), we test for increased interest in vasectomies using new patient consultations and Google trends. Appendix A4 shows no evidence of increased interest in vasectomies.

restrictions note that the burden of these policies fall unequally on vulnerable populations, potentially furthering societal inequities (Fuentes, 2023).

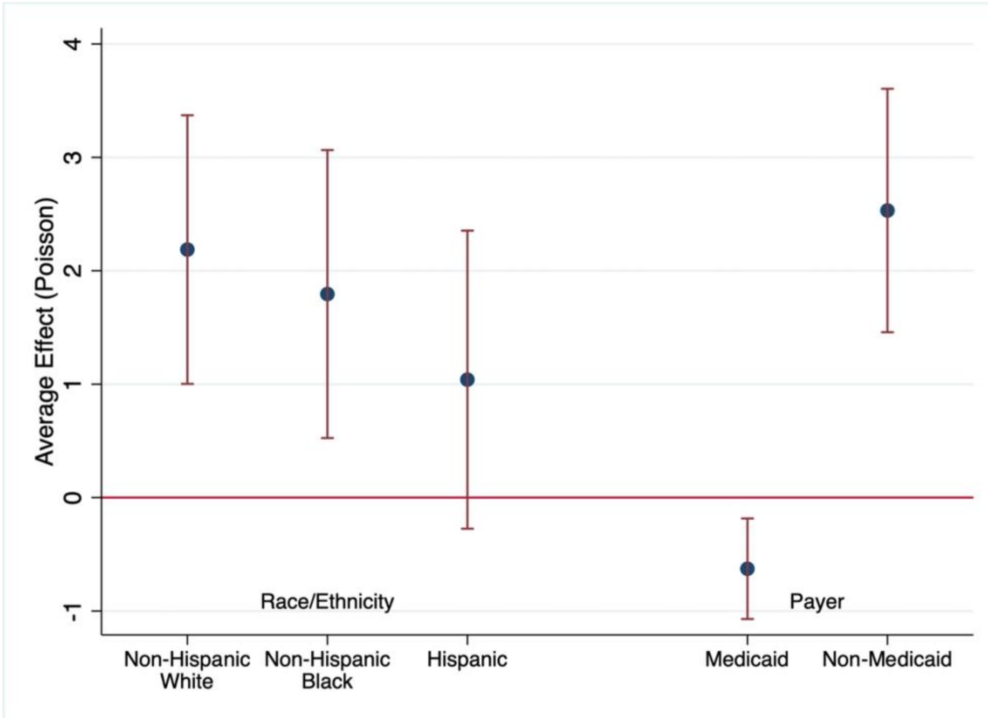
To consider heterogeneous responses to the IUD use in responses to HB2, we estimate the following specification:

$$E[Y_{ct} \mid \alpha_c, \delta_t, treat\_period] = \exp(\alpha_c + \delta_t + \beta(treat\_period)) \qquad (2)$$

Where Y_{ct} is the number of IUD insertions in county c at quarter-year t and *treat_period*=1 for treated counties in the 6 quarters surrounding the passage of HB2, equivalent to the quarter-years where we see effects of the policy among the total population of reproductive age women. We estimate the effects separately by racial/ethnic group including: Non-Hispanic white; Non-Hispanic Black; Hispanic; and primary insurance, including Medicaid and Non-Medicaid.

Figure 6 shows the results when we estimate Equation (2) separately by race and ethnicity. The change in IUD rates among Non-Hispanic white women is roughly double the change for Hispanic women. Additionally, the magnitude and direction of the effects are different across primary insurance. IUD rates increase among those not using Medicaid, while the rate of IUD insertions paid for by Medicaid slightly declines.

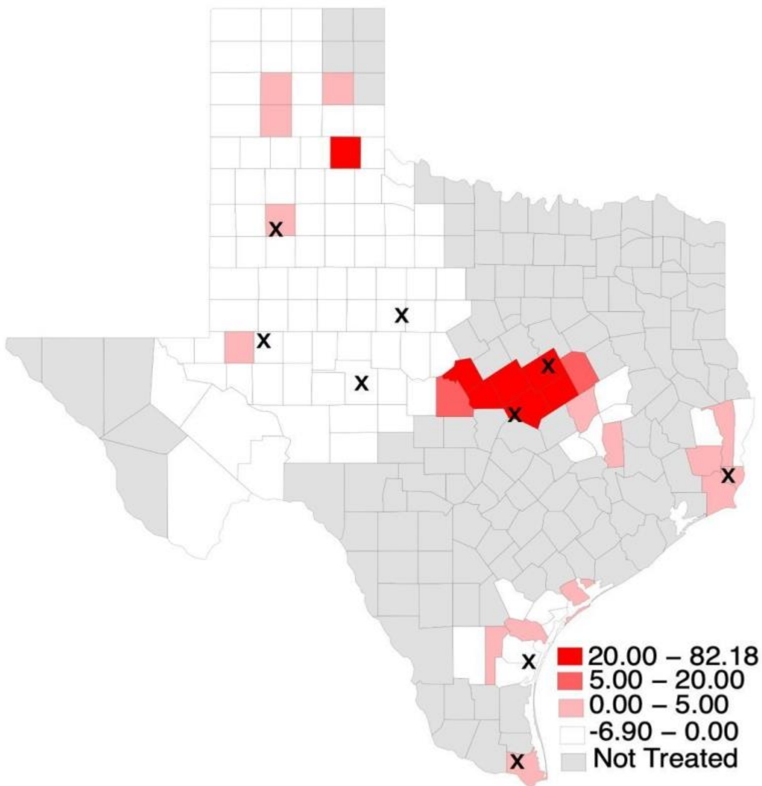
Figure 6: Heterogeneity of HB2 Effects by Race/Ethnicity and Payer



Notes: The graph shows the estimates from Equation (2) estimated separately by racial/ethnic group and payer. In the racial analyses, the Poisson specification includes an exposure variable equal to the population of reproductive age females in the corresponding racial/ethnic group. In the analysis by primary payer, the exposure variable is equal to the total population of reproductive age females.

We also analyze rural and suburban differences in IUD insertion rates. Figure 7 is a heat map that describes the 2011-2013 change in the rate of IUD insertion per 10,000 resident females in counties treated by HB2. The largest changes in the rate of IUD insertion following HB2 occur in counties surrounding the metro areas of Waco and Houston, as well as in some of the more densely populated areas of West Texas, Amarillo, and Lubbock. These areas are largely suburban, and therefore the populations that appear to be responding to the change in abortion access are concentrated in suburban counties, rather than in rural areas of the state. In particular, counties that respond the most are often near the locations of clinic closures following HB2 (Ura et al., 2016). The dark red cluster in the center of the state represents counties in the immediate vicinity of Planned Parenthood clinics in Waco and a women's health clinic in Killeen. There are also counties with high IUD insertion rates after HB2 implementation near clinic closures in Beaumont, McAllen, and Corpus Christi.

Figure 7: Change in IUD Insertion Rates Among Treated Counties



Notes: The heat map above shows the 2011 to 2013 change in the rate of LARC insertion per 10,000 resident females in counties that are treated by HB2, with darker red represented a larger change in insertions. X's represent clinic closures identified by Ura et al. (2016).

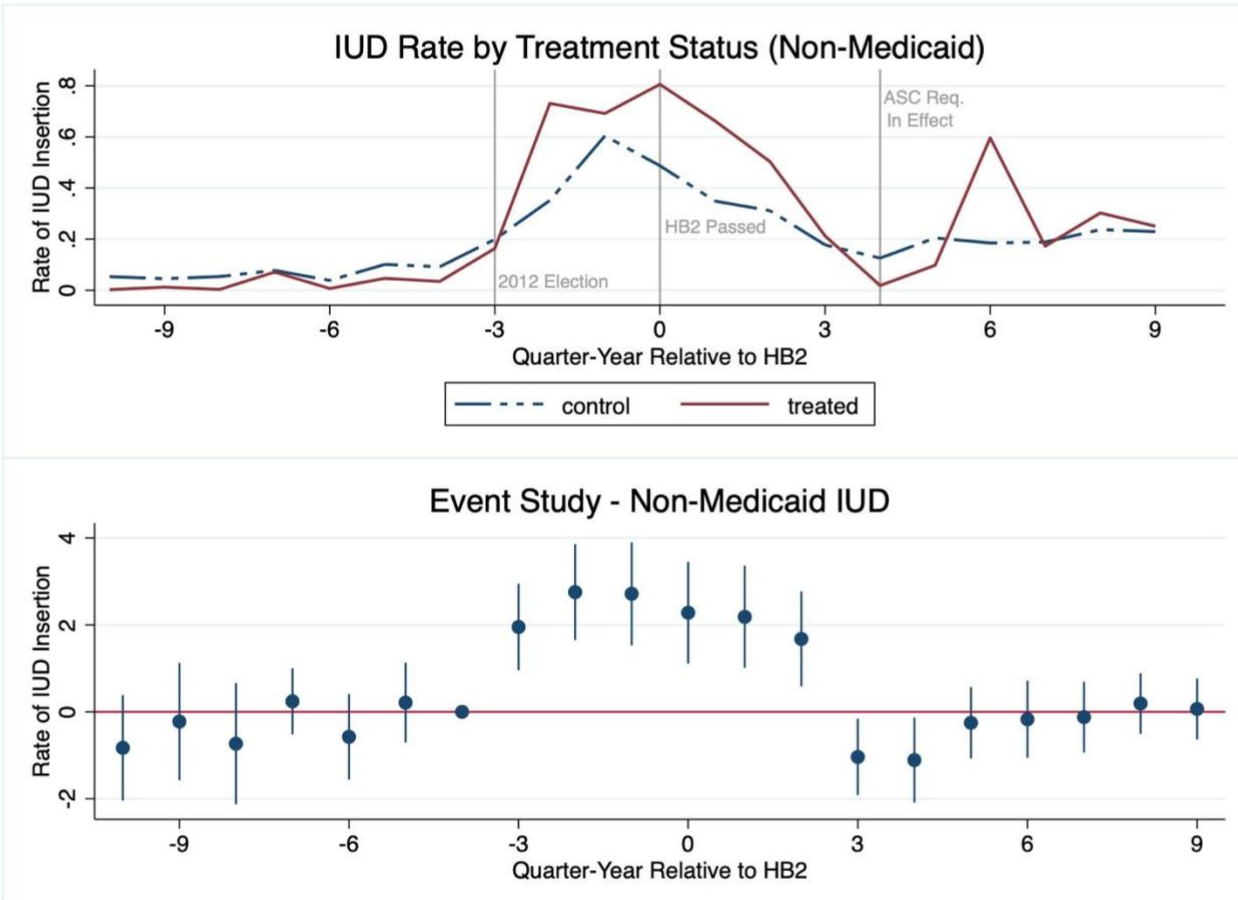
The differences in responding and non-responding populations reflect known disparities in women's health, which is an area of concern for the American College of Obstetricians and Gynecologists (American College of Obstetricians and Gynecologists, n.d.), a professional membership organization for obstetricians and gynecologists. Disentangling the mechanisms driving the differences in IUD choice post-HB2 is an area for future research, though descriptive analysis suggests that there is variation in IUD use that may deepen disparities in women's health.

4.3 Exclusion of Medicaid recipients

While we observe IUD insertions and vasectomies in hospital-based outpatient and ASC settings, one limitation of the data is that we do not observe procedures in facilities such as healthcare clinics without a hospital affiliation or community health centers. Therefore, we are concerned that the observed trends are driven by compositional changes in hospital choice in Texas during this time period. As described in Section 2.2, Texas excluded sexual health clinics that are affiliated with or make referrals to abortion providers from receiving Medicaid reimbursement. At the same time, Texas dramatically cut its funding for family planning clinics. Lastly, many abortion clinics that also provide family planning services may have closed due to the implementation of HB2 in mid-2013. All three policies could cause people to switch from care in independent clinics toward hospital-owned facilities, resulting in a false conclusion that take-up of these contraceptive methods increased in the state overall.

One way we address this possibility is by repeating the estimation of IUD insertions while only considering non-Medicaid cases. The decision to exclude family planning clinics from receiving reimbursement from the Texas Women's Health Program and the budget cuts that led to dramatic family planning clinic shutdowns could drive Medicaid recipients from independent family planning clinics, such as Planned Parenthood affiliates, toward hospital-based outpatient facilities. If this facility shift occurs primarily in the group of counties we identify as treated by HB2, it will result in increases in IUDs in our data that we falsely attribute to restricted abortion access. We repeat the event study analysis for IUD insertions in Equation (1) excluding Medicaid recipients from our sample.

Figure 8: Trends and Event Study – Non-Medicaid IUD



Notes: The top panel of this figure shows trends in IUD procedures, excluding individuals who were covered by Medicaid. The bottom panel shows the λ_k coefficients from Equation (1) using only individuals who received an IUD and did not have Medicaid.

Figure 8 shows that excluding Medicaid recipients does not fundamentally shift the trends between treatment and control counties, or the estimates shown in the event plot, providing evidence that Medicaid recipients are not driving the changes in IUD insertion around the timing of HB2. This supports a conclusion that the family planning clinic budget cuts and changes to the

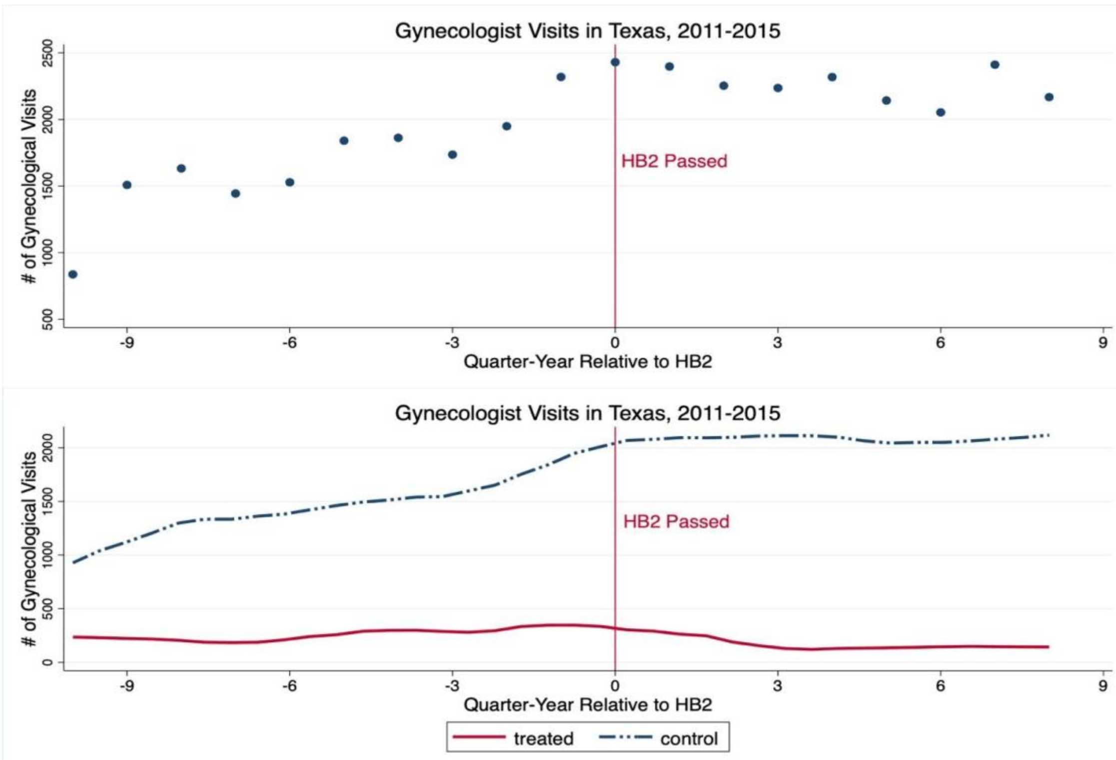
Texas Women's Health Program are unlikely to cause the increases in IUD insertions we observe for counties that experience increased distance to an abortion provider following HB2. Fischer et al., (2018) show that there is little correlation between increased distance to an abortion provider following HB2 and restricted access to publicly funded family planning clinics after the change to the Texas Women's Health Program.

4.4 Gynecological Visits

Policies that cause abortion clinics to close may also reduce access to all other services provided by the clinics, such as general reproductive health checkups, contraception, prenatal care, and cancer screenings. Abortion clinic closures may also cause individuals to switch their reproductive healthcare to another facility after HB2. For example, consider the following scenario: an individual regularly seeks care at a small family planning clinic, not included in the THCIC claims data. As a result of HB2, the clinic closes and the individual switches to a hospital-owned facility in the THCIC data. As a result, we observe a mechanical increase in IUD rates and falsely attribute this change to HB2. While this response is a consequence of HB2, it is not necessarily related to restricted abortion access. Therefore, this behavior may be confounding our central research question.

We explore the possibility that people in areas experiencing facility closures shift their care to a hospital-owned facility or ASC in our data by observing trends in other reproductive health services. In Figure 9, we plot the average trend and the trends by treatment status for women's health screenings over the study period. If people are making a fundamental shift in their preferred facility for care, we expect corresponding increases in visits to the gynecologist in treated counties after clinic closures due to HB2. Figure 9 shows that the overall incidence of women's health screenings did increase over the study period, but this increase occurs almost exclusively in the control group. Among treated counties, the incidence of women's health screenings is stable. So, it does not appear that residents of counties affected by HB2 systematically shift to receive healthcare in hospital-owned and ASC facilities.

Figure 9: Trends in General Gynecologist Visits



Notes: The top figure shows general trends in visits to the gynecologist from 2011-2015. Each blue dot represents the county of visits in each quarter-year, and the vertical red line represents July 2013, when HB2 was passed. The bottom figure shows the number of gynecologist visits in the treated counties, represented by the blue trend line, and control counties, represented by red trend line. Like the figure above, the vertical red line represents June 2013.

5. Conclusion

In this paper, we measure the effect of restricted abortion access in Texas on the incidence of hospital-based IUDs and vasectomies. For identification, we exploit within-state geographic variation in the distance to the nearest abortion provider after the passage of HB2, which shut down over half of all abortion clinics in Texas. We find that counties with an increase in their travel distance to an abortion provider greater than 30 miles experience an average increase of 0.55 IUD insertions per 10,000 reproductive age females in treated counties around the time of the policy change. We do not find evidence that counties affected by HB2 experience increases in vasectomies.

A primary threat to identification is that our data on outpatient procedures only includes discharge records for IUDs and vasectomies that occurred in a hospital-owned facility or ambulatory surgical center. As such, our results may be biased by changes in the location of care resulting from these policies. However, we argue that the public policy environment in Texas did not significantly shift the location of care in the state from independent clinics toward hospital-owned facilities and conduct additional tests to ensure the robustness of our results. Our results are robust to the exclusion of Medicaid recipients from the sample, a population that experienced reduced access to contraceptive care in publicly funded clinics after changes to the reimbursement structure for the state Women's Health Program in 2012. In addition, we do not find evidence that counties affected by the abortion clinic closures following HB2 increased the number of hospital-based general gynecologist visits, supporting a conclusion that individuals in these counties did not make large changes in their location of reproductive healthcare during the study period.

Our study has a few key limitations. When presented with new information, people may respond to changes in their expectation of abortion access in the future, regardless of realized restrictions in access. In this way, people across the entire state of Texas may be influenced by the media surrounding abortion access during the 83rd legislative session, and therefore defining a treatment group to only include counties affected by abortion clinic closures may be too narrow. So, we consider our treatment effects to be conservative estimates of changing contraceptive behavior. Additionally, we only measure the effects of abortion access on contraceptive options that require an outpatient visit to a physician. According to the Guttmacher Institute (2021) LARC is the third most common contraceptive method among contraceptive users aged 15-48, behind sterilization (27.7 percent) and the pill (21.4 percent). More research is necessary to determine the influence of abortion access on the take-up of all types of contraception, rather than IUDs alone. Finally, our study relies on variation in abortion access in a single US state. Although Texas is populous, it contains only 8.9 percent of the total US population, and results may not be generalizable to the entire country.

To our knowledge, our study is the first in the United States to provide empirical evidence supporting the hypothesis that abortion and IUDs are substitutes, making two contributions to the broader conversation on the impact of abortion restrictions. First, our work suggests that individuals engage in defensive contraception when faced with abortion restrictions, with larger increases in areas most affected by the restrictions. In the context of the *Dobbs v. Jackson Women's Health Organization* decision, we expect individuals to increase IUD utilization, particularly in the thirteen 'trigger' states. So far, medical research by Ellison et al. (2024) finds increases in permanent contraception procedures, including both tubal ligation and vasectomy, after *Dobbs*. Although we do not find changes in vasectomy procedures, it is possible that individuals chose permanent contraceptive methods in response to the stronger abortion restrictions imposed by *Dobbs*.

In addition, research on abortion restrictions consistently finds increases in birth rates (Jones & Pineda-Torres, 2024; Venator & Fletcher, 2021; Myers & Ladd, 2020; Guldi, 2008; Kane & Staiger, 1996). However, if a subset of the population chooses a more effective contraception option in response to abortion restrictions, the effect of these restrictions on birth rates identified in prior literature may be a lower bound. If the cost of accessing IUDs or other contraceptive methods increased alongside abortion restrictions, we may expect larger increases in birth rates than previously estimated.

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Appendix

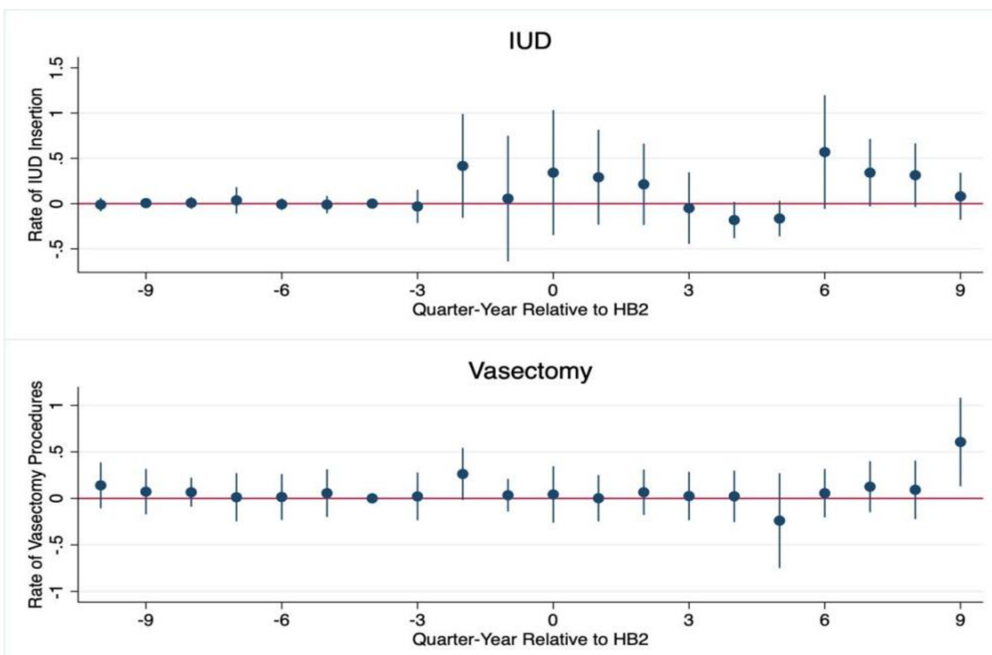
A.1 Linear specification

We repeat our main event-study specification using a linear model, rather than a Poisson. The model takes the following form:

$$Y_{ct} = \alpha_c + \delta_t + \sum_{k=-10}^{9} \lambda_k 1(t = k) + \epsilon_{ct} \qquad \text{(A1)}$$

Y_{ct} is the rate of IUDs or vasectomies in county c at quarter-year t, and α_c and δ_t are county and quarter-year fixed effects respectively. Again, k indexes the coefficients λ according to the time relative to the passage of HB2 in 2013 quarter 3. Figure A1 displays the results of the linear specification.

Figure A1: Linear event studies

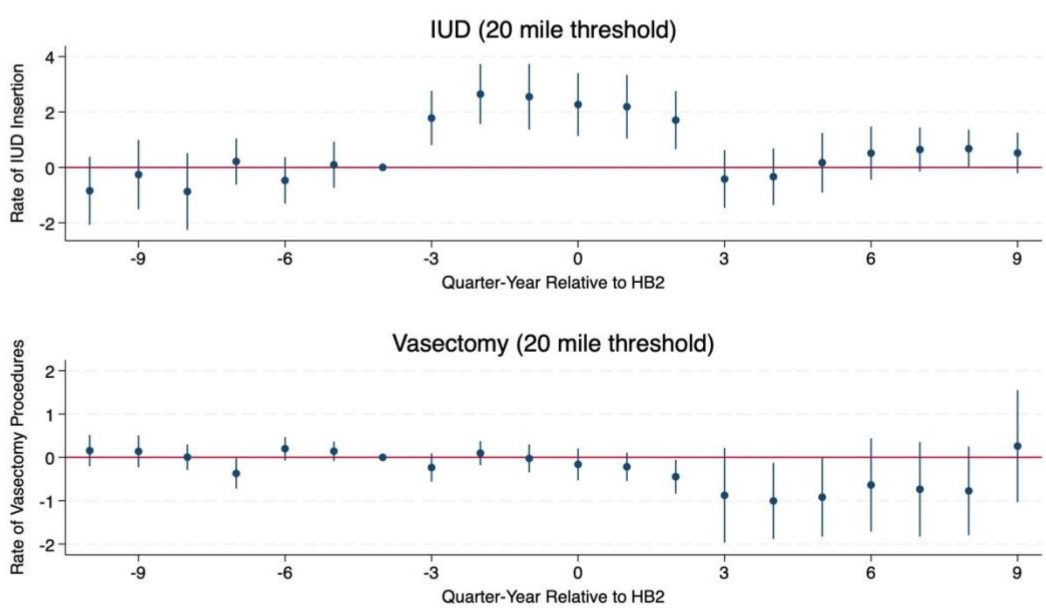


Notes: The figure shows the results of a linear specification that estimates the change in the rate IUD insertions (top) or vasectomies (bottom) in the treated counties relative to the control counties after the introduction of HB2.

A.2 Sensitivity to Treatment Definition

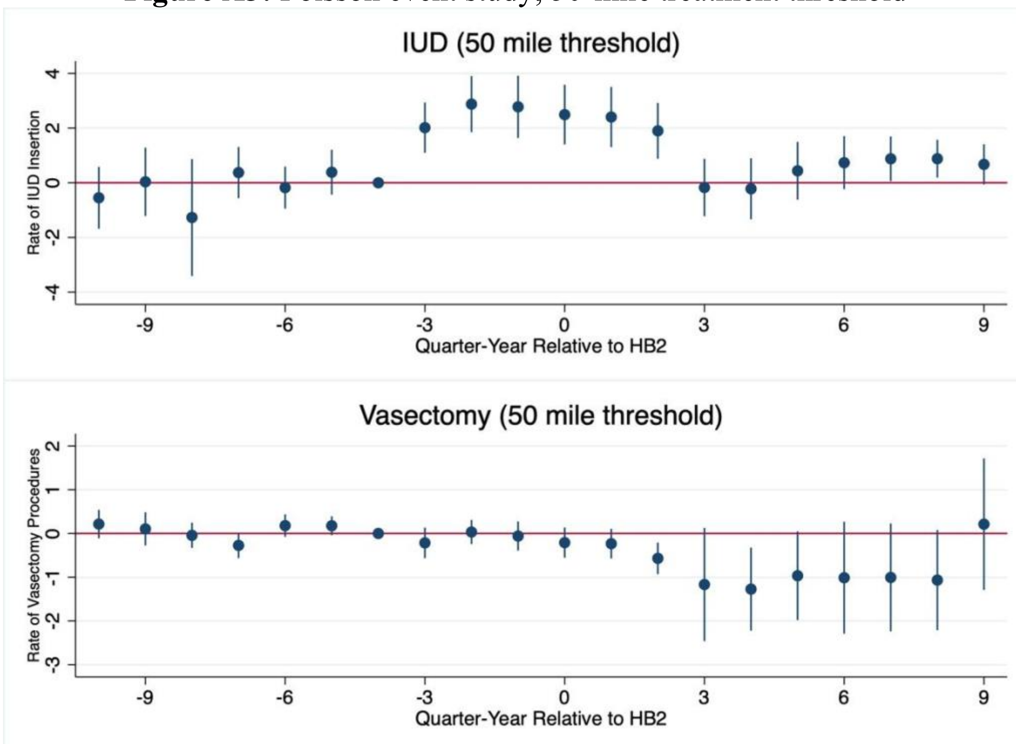
We test for the sensitivity of our analysis to different treatment definitions. Instead of our primary treatment definition of a change in the distance to the nearest abortion provider greater than 30 miles, we look at increases in the distance of more than 20 miles, more than 50 miles, and more than 100 miles. Figure A2 shows the results of the event study analysis using a 20-mile treatment threshold, equivalent to the median distance change following HB2. Figure A3 shows the results using a 50-mile threshold. For the results presented in Figure A3, we exclude counties that experience an increase between 30-49 miles. This allows us to limit the bias that would be introduced if we considered these counties as part of the control group. The results of the analyses in Figure A2 and A3 are consistent with the greater than 30-mile threshold, suggesting that the results are not sensitive to small changes to the distance threshold for treatment in either direction.

Figure A2: Poisson event study; 20-mile treatment threshold



Notes: Similar to Figure 5, this figure plots the estimates from Equation (1) but uses a 20-mile treatment threshold.

Figure A3: Poisson event study; 50-mile treatment threshold

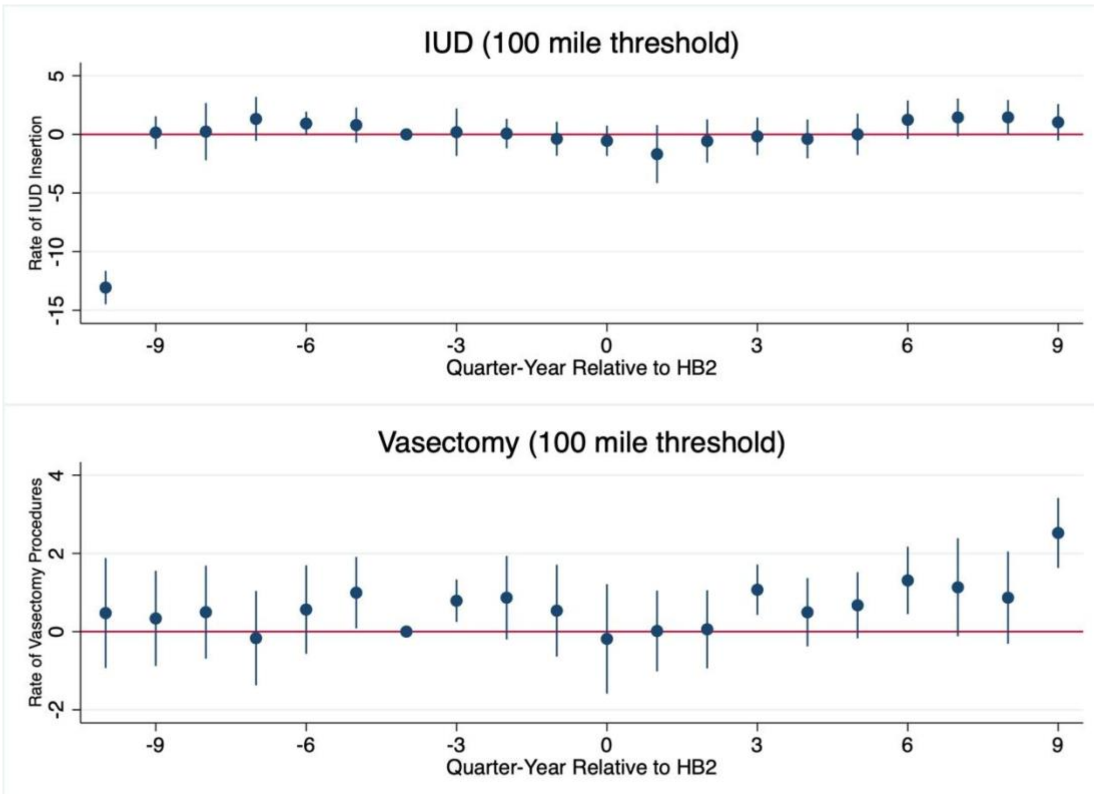


Notes: Similar to Figure 5, this figure plots the estimates from Equation (1) but uses a 50-mile treatment threshold. These estimates exclude counties with an increase of 30-49 miles.

Figure A4 shows the results of an event study using an increase of greater than 100 miles as the threshold distance. As with the greater than 50-mile threshold estimation used to create Figure A3, we exclude individuals who experience an increase of 30-99 miles to the nearest abortion provider.

The event plots are quite different under the 100-mile threshold and are much noisier. Of note, 86 percent of our response group come from just 3 suburban counties around Waco. When we limit our analysis to just those counties who experience a greater than 100-mile increase, these suburban counties are no longer included in the analysis. Additionally, LARC activity is rare in counties that experience a greater than 100-mile increase in distance. For example, the average number of IUD insertions is 1.08 per county-quarter, but when we limit to only counties experiencing a greater than 100-mile increase in distance from an abortion provider, the average number of IUD insertions is 0.13 per county-quarter. This could explain why the event studies for the greater than 100-mile treatment group are quite noisy.

Figure A4: Poisson event study; 100-mile treatment threshold



Notes: Similar to Figure 5, this figure plots the estimates from Equation (1) but uses a 100-mile treatment threshold. These estimates exclude counties with an increase of 30-99 miles.

A.3 County Controls

In this section, we repeat our primary specification and include county-level control variables. Table A1 presents county-population-level summary statistics for treated and untreated counties. The treatment definition for this specification is the same as the primary specification; counties are treated if the distance to an abortion provider increased more than 30 miles following HB2.

Treated counties have smaller average populations than untreated counties, along with a slightly higher proportion of residents who identify as white or Hispanic, and a slightly lower proportion of residents who identify as Black. The proportion of the total population that is within the age range of 20-49 is similar between both treated and untreated counties. Finally, treated counties have a lower unemployment rate and median household income, but higher income per capita.

Table A1: Summary Statistics for Treated and Untreated Counties

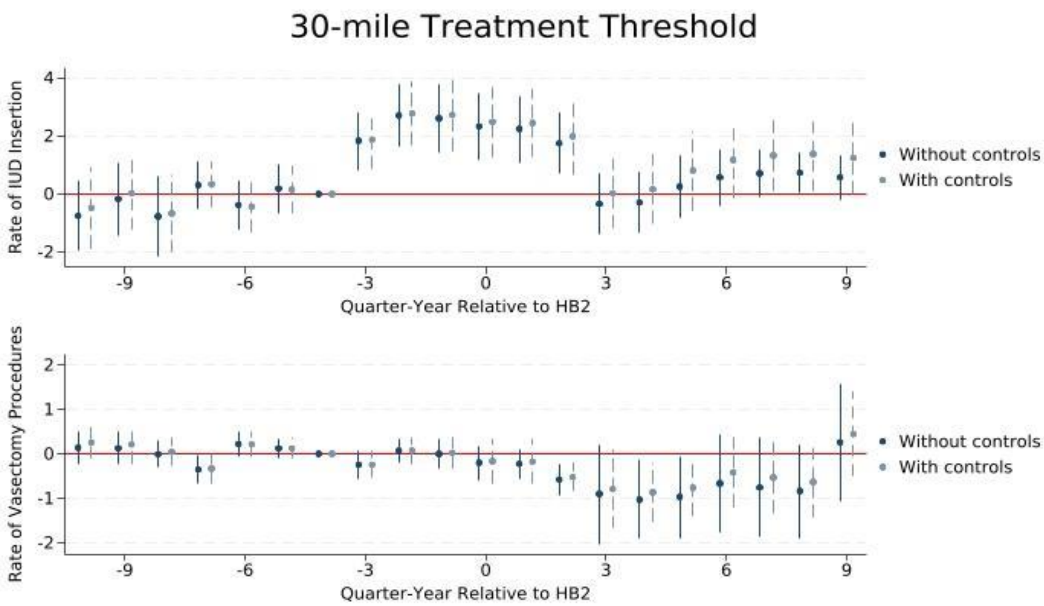
	Treated		Untreated	
Variable	Mean	County-Years	Mean	County-Years
County Population	36,479	2,338	163,756	2,740
Race (%)				
White	92.11		89.81	
Black	6.27		8.25	
Hispanic	36.14		31.68	
Gender (%)				
Female 20-29	5.67		5.79	
Female 30-39	5.49		5.65	
Female 40-49	5.71		6.09	
Male 20-29	7.29		6.53	
Male 30-39	6.46		5.94	
Male 40-49	6.22		6.21	
Employment and Income				

Unemployment Rate	5.43	6.19
Poverty Rate	17.79	17.74
Per capita Personal Income	$41,741.37	$39,697.25
Median Household Income	$44,628.00	$47,148.09

Source: U.S. Census Bureau; Bureau of Labor Statistics 2010-2016

The primary specification is robust to including county controls. In the Figure A5, we include the estimates from Figure 5 and superimpose estimates that include the variables in Table A1 as county controls. The original estimates are in solid blue and the estimates from the specification including county controls are represented by dashed lines. The estimates maintain the same trend but do become less precise. While not shown, this holds true when the treatment is defined at 50 miles and at 100 miles.

Figure A5: Poisson event study with and without control variables

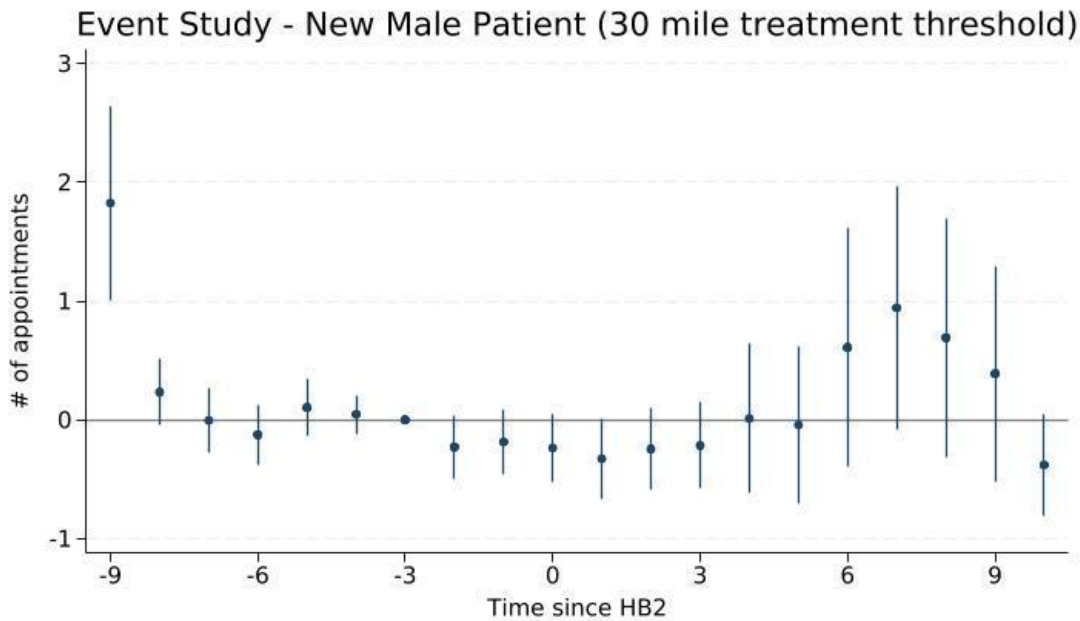


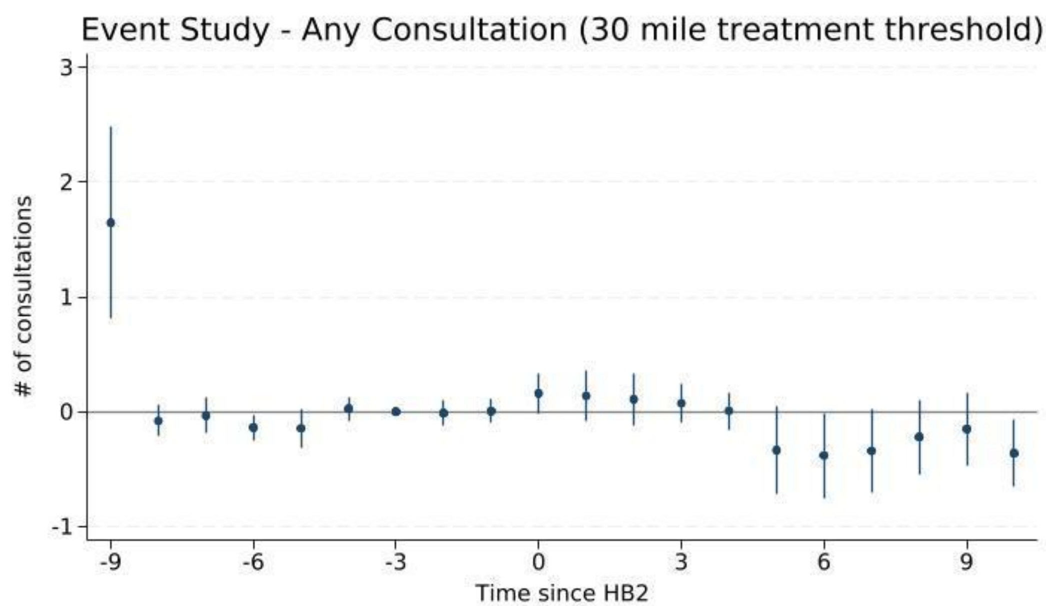
Notes: The event studies plot both the estimates plotted in Figure 5 (solid line), which do not include control variables and estimates that include a set of controls (dashed line).

A.4 Vasectomy

While we do not observe a shift in vasectomy procedures following HB2, there may be increased interest in the procedure. We test whether this is the case using two methods. First, we look at the change in new patient consultations. Vasectomy procedures usually begin with a consultation, where patients meet with a physician to discuss the risks and benefits of the procedure. It is possible that individuals consider getting a vasectomy in response to abortion restrictions and change their mind after the consultation. Though our data does not specifically identify patient visits where a vasectomy is discussed, we look broadly at male patients with any new patient visit and specifically at male patients with a visit coded as an encounter for either contraceptive management (ICD-9 V25 or ICD-10 Z30) or procreative management (ICD-9 V26 or ICD-10 Z31) but not for a vasectomy (V25.2 or Z30.2). We estimate the change visits in response to HB2 using Equation (1) with the count of new patient visits or reproductive consultations as the dependent variables and present the results in Figure A6. The number of new patient visits in treated counties does not significantly differ after the treatment date relative to the control counties. We take this as evidence that male patients are not seeking out office visits to discuss having a vasectomy.

Figure A6: Poisson event study; new patient visits

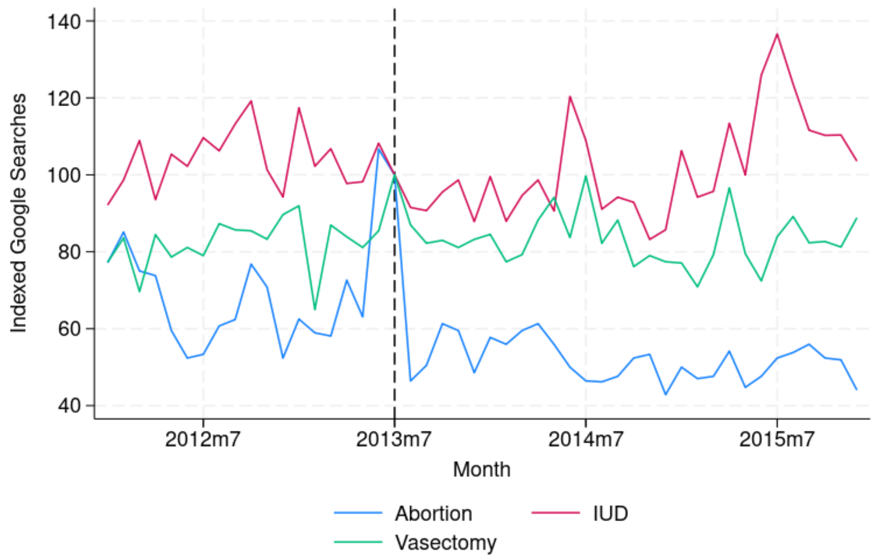




Notes: This figure uses Equation 1 to estimate the change in new patient visits (top) and reproductive consults (bottom) for male patients. The estimates show the change in appointments relative to the control counties after HB2, represented by zero.

Next, we measure interest in vasectomy through Google search trends for the keywords "vasectomy," "IUD," and "abortion" in Texas from January 1, 2012, to December 31, 2015. The trends are measured relative to the week that the search was most popular. We collapse the search history to the monthly level and index the search intensity to July 2013, which gives the relative search intensity compared to the month that HB2 was passed. Values above 100 and below 100 indicate that the popularity of the searched term was higher or lower, respectively, than the popularity in July 2013. The trends are shown in Figure A7. Google searches for vasectomies peaked in July of 2013 and remained relatively stable. Searches for abortions peaked in June of 2013 and remained high for July but decreased by roughly 40 points afterwards. Last, searches for IUDs peaked in July of 2015, with July of 2013 not exhibiting different search intensity than the surrounding months.

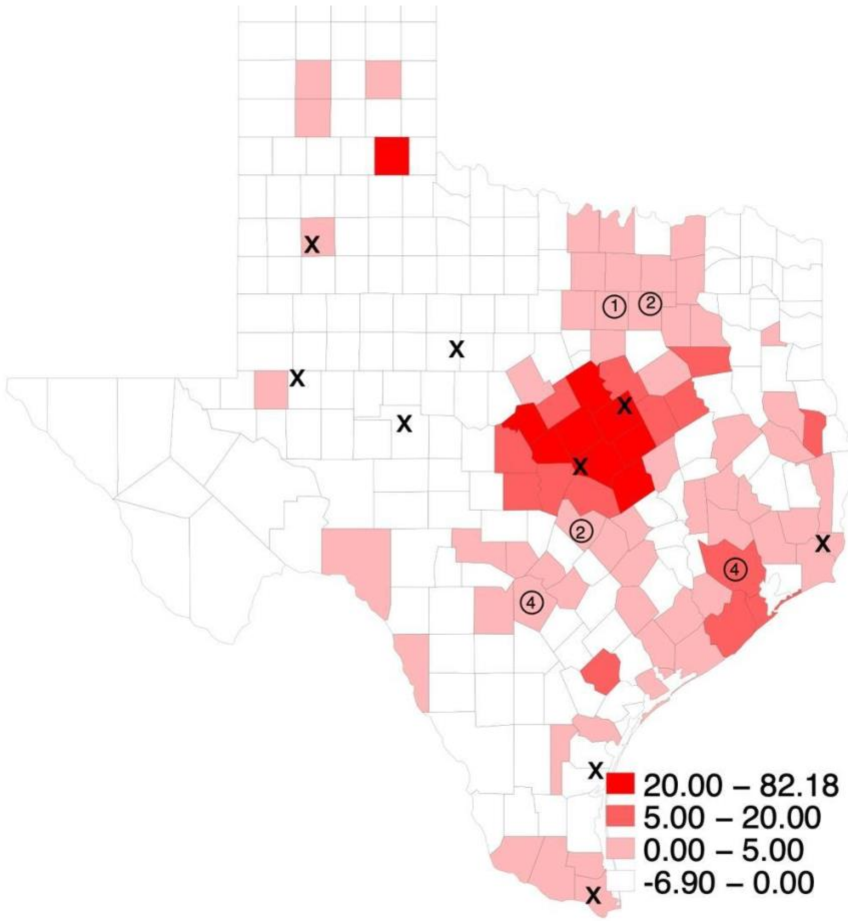
Figure A7: Google searches relative to July 2013



Notes: In the figure above, we plot the Google trends for the vasectomy (green), abortion (blue), and IUD (pink) relative to trends in July 2013. Values above 100 indicate that searches were higher than July 2013, while values below 100 indicate that searches were lower than July 2013.

A.5 Information Saliency

Figure A8: Changes in the rate of IUD insertion, 2011-2013 (all counties)



Notes: The heat map above shows the 2011 to 2013 change in the rate of IUD insertion per 10,000 resident females in all counties. X's represent areas that lost all clinics, and circles represent areas that lost access to some, but not all, clinics. The number inside the circle indicates the number of clinic closures. Clinic closures are identified by Ura et al. (2016).

Given the strength of the anticipation effects that we observe, we consider a potential alternative treatment definition in this section. If increases in contraceptive behavior are defensive measures in response to restricted abortion access, it may be the saliency of the information surrounding the effects of HB2 that matters in contraceptive decision making, rather than the eventual increase in travel distance. Figure A8 is a heat map similar to Figure 6 displaying the increase in IUD insertion per 10,000 resident females. However, unlike Figure 6, Figure A8 includes all counties. The X's represent counties that lose all abortion providers, while the circles represent counties that lose some but not all providers. The number in the center of the circle indicates the number of clinics that closed following HB2. Figure A8 suggests that residents of metro areas that lost some, but not all, abortion providers moderately increase their use of IUDs. There are small increases in the rate of IUDs in Dallas-Fort Worth, Austin, and San Antonio, and a slightly larger increase in IUD rates in Houston. While all of these regions lost between one and four abortion providers, individuals residing in these areas do not experience large changes in travel distance since not all providers closed. Our paper argues that the increase in demand for IUDs is driven by individuals facing the loss of all abortion providers in their area. However, a competing narrative is that exposure to any clinic closure increases the saliency of the abortion restrictions imposed by HB2, and therefore individuals faced with the loss of any abortion provider have an incentive to invest in defensive contraception.

To explore the plausibility of this narrative, we repeat our event study approach using a new definition of treatment. In this setting, a county is treated when the population centroid of the county is within 50 miles of the location of an abortion clinic closure.[8] All counties outside the 50-mile radius of the closure are considered control counties, regardless of changes in travel distance. Figure A9 provides the results of this specification, which is equivalent to Equation (1) using the new treatment definition. The event study shows that rates of IUD insertion increase modestly around the timing of HB2, but the effects are often not statistically significant. In this analysis, there is limited evidence of any anticipation effects before the policy goes into effect.

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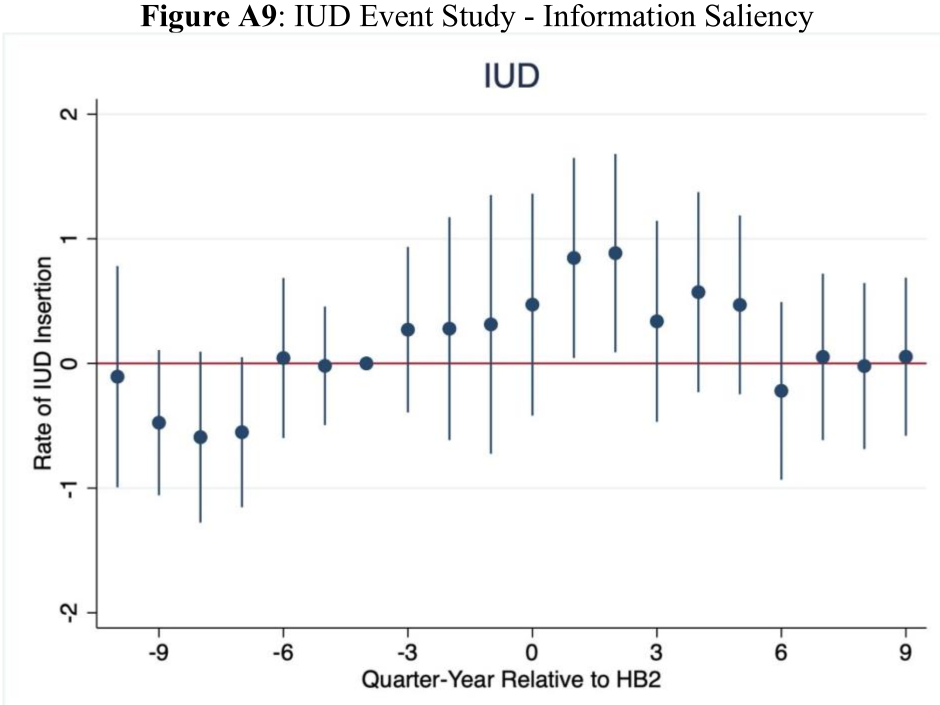
[8] We perform this exercise using the addresses of the clinic closures following HB2, population centroids from the Census, and the georoute command in Stata to calculate distances.

Instead, the strongest effects occur in the two quarters immediately following the passage of the law. So, this may be evidence that people living close to these abortion providers only change their contraceptive behavior after observing the closure.

One important limitation of this approach is that it generates a very stark urban/rural divide across the treatment and control groups. Because abortion clinics in Texas are primarily located in urban areas, defining treatment based on the proximity to these clinics will result in a treatment group of primarily urban counties and a control group of primarily rural counties. The trends in IUD insertion rates in rural counties may not serve as a reasonable control for trends in urban counties. This could also explain why the event study in Figure A9 shows some evidence of pre-trend differences, which may confound the results.

Figure A9: IUD Event Study - Information Saliency



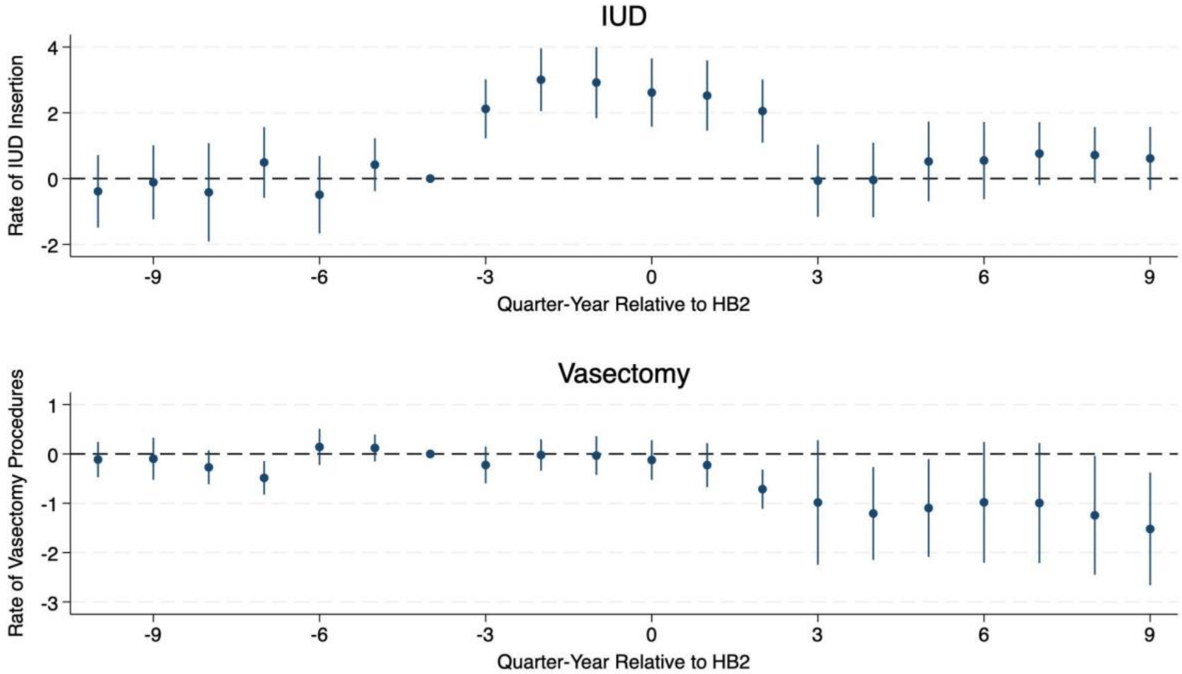
Notes: Figure A9 provides estimates from an event study specification using Equation (1) with a treatment definition that depends on the county-level distance from an abortion clinic closure. Counties less than 50 miles from the location of a clinic closure following HB2 are considered treated, while all other counties are in the control group. 90% confidence intervals are shown.

A.6 Rural Hospital Closures

In the past few decades, a significant number of rural hospitals closed or stopped providing certain services due to financial instability (Economic Research Service, 2025). If the rapid closing of rural hospitals in Texas limited access to contraceptive services, this may confound the estimates in our primary specification. In particular, our treatment group includes a large portion of rural west Texas, and closures in this region may cause our primary results to underestimate the effects of HB2 if the law coincides with reduced access to contraceptive services.

To ensure that our results are not biased by hospital closures, we repeat our primary specification using a balanced sample of 578 unique facilities. In the outpatient, THCIC data, facilities come in and out of the sample quarter-to-quarter due to a variety of reasons, including closures and the standards of reporting requirements. Figure A10 shows the event study results from Equation (1) using only data from the balanced sample of facilities. These results are identical to our primary specification shown in Figure 5, demonstrating that our results are robust to this sample limitation.

Figure A10: Poisson Event Study - Balanced Sample of Facilities



Notes: Figure provides the event study estimates from Equation (1) using data only from a balanced sample of facilities in the outpatient data. 95% confidence intervals are shown.